SAMPLE

Solar System Missions: Venera

Soviet Probes to Venus

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Introduction

The Venera program (Russian: ???????, pronounced [v???n??r?] 'Venus') was a series of space probes developed by the Soviet Union between 1961 and 1984 to gather information about the planet Venus. A total of eighteen probes were sent, including two related Vega probes. Thirteen probes successfully entered the Venusian atmosphere, including the two Venera-Halley probes. Ten of those successfully landed on the surface of the planet.
Due to the extreme conditions, the probes could only survive for a short period on the surface, from 23 minutes to two hours.
The Venera program established a number of precedents in space exploration, among them being the first man-made devices to enter the atmosphere of another planet (Venera 3 on 1 March 1966), the first to make a soft landing on another planet (Venera 7 on 15 December 1970), the first to return images from another planet's surface (Venera 9 on 8 June 1975), the first to record sounds on another planet (Venera 13 on 30 October 1981), and the first to perform high-resolution radar mapping scans (Venera 15 on 2 June 1983).

The program that follows in these pages is therefore two stories at once. One is a scientific detective story: how a featureless white disk in telescopes turned out to be a furnace world with crushing pressure, acid clouds, and a geology rewritten by volcanism, a place that reset every assumption about what a planet like Earth could become. The other is an engineering saga: how designers with slide rules and test chambers built machines that could not only reach that world but work on it, again and again, each flight fixing what the last one taught. Both stories are worth the telling, and neither is complete without the other.
The intense pressures that led to the short lifespans of many of the probes was itself an important finding. The first Soviet attempt at a flyby probe to Venus was launched on 4 February 1961, but failed to leave Earth orbit. In keeping with the Soviet policy at that time of not announcing details of failed missions, the launch was announced under the name Tyazhely Sputnik ("Heavy Satellite"). It is also known as Venera 1VA.
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Venera 14 (called Venus 14 in English) was a probe in the Soviet Venera program for the exploration of Venus.
The list of lander experiments and instruments include:
Thermometers, Barometers - ITD
Spectrometer / Directional Photometer - IOAV-2
Mass spectrometer - MKh-6411
Penetrometer / Soil ohmmeter - PrOP-V
Chemical Redox indicator - Kontrast
2 color telephotometer cameras - TFZL-077
Gas chromatograph - Sigma-2
Radio / Microphone / Seismometer - Groza-2
X-Ray Fluorescence Spectrometer (Aerosol) - BDRA-1V
X-Ray Fluorescence Spectrometer (Soil) - Arakhis-2
Soil Drilling Apparatus - GZU VB-02
Stabilized Oscillator / Doppler Radio
Small solar batteries - MSB
Venera 14 landed at 13.25°S 310°E / -13.25; 310, about 950 kilometres (590 mi) southwest of Venera 13, near the eastern flank of Phoebe Regio on a basaltic plain.
Venera 13 (Russian: ??????-13 'Venus 13') was part of the Soviet Venera program meant to explore Venus.
Venera 13 and 14 were identical spacecraft designed for the 1981 Venus launch window. The probes were launched five days apart, with Venera 13 launching on 30 October 1981 at 06:04 UTC and Venera 14 launching on 4 November 1981 at 05:31 UTC. Both carried an on-orbit dry mass of 760 kg.
Venera 13 transmitted the first recording of sounds from another planet, including sounds of Venusian wind, the lander hitting the ground, pyrotechnic lens cap removal and its impact on regolith, and action of the regolith drilling apparatus. The descent lasted for about an hour. Venera 13 landed on 1 March 1982 at 03:57:21 UT at 7.5 S, 303 E, just east of the eastern extension of an elevated region known as Phoebe Regio.
The area is composed of bedrock outcrops surrounded by dark, fine-grained soil.
As the cruise stage flew by Venus, the bus acted as a data relay for the lander before continuing on to a heliocentric orbit. The probe was equipped with a gamma-ray spectrometer, UV grating monochromator, electron

and proton spectrometers, gamma-ray burst detectors, solar wind plasma detectors, and two-frequency transmitters which made measurements before, during, and after the Venus flyby. The bus continued to provide data until at least 25 April 1983.
The lander was a hermetically-sealed pressure vessel and contained most of the instrumentation and electronics. The lander survived the heat by containing blocks of lithium nitrate, a salt-like substance which has a very high heat of fusion. Mounted on a ring-shaped landing platform and topped by an antenna, the landers design was similar to the earlier Venera 9-12 landers.
An air brake was used to slow down Venera, (this is different to the previous landers, which used parachutes, making them prone to failure.).
On 1 March 1982, after launch and a four-month cruise to Venus, the descent vehicle separated from the cruise stage and plunged into the Venusian atmosphere.

Chapter 1: The Discovery of Venera

Venera 2 launched on 12 November 1965, but also suffered a telemetry failure after leaving Earth orbit.
Several other failed attempts at Venus flyby probes were launched by the Soviet Union in the early 1960s, but were not announced as planetary missions at the time, and hence did not officially receive the "Venera" designation. The Venera 3 to 6 probes were similar in construction. Weighing approximately one ton, and launched by the Molniya-type booster rocket, they included a cruise "bus" and a spherical atmospheric entry probe.
The probes were optimised for atmospheric measurements, but not equipped with any special landing apparatus. Although it was hoped they would reach the surface still functioning, the first probes failed almost immediately, thereby disabling data transmission to Earth. Venera 3 became the first human-made object to impact another planet's surface as it crash-landed on 1 March 1966. However, as the spacecraft's data probes had failed upon atmospheric penetration, no data from within the Venusian atmosphere were retrieved from the mission.
On 18 October 1967, Venera 4 became the first spacecraft to measure the atmosphere of another planet.
This spacecraft first showed the major gas of Venus's atmosphere to be CO2. While the Soviet Union initially claimed the craft reached the surface intact, re-analysis, including atmospheric occultation data from the American Mariner 5 spacecraft that flew by Venus the day after its arrival, demonstrated that Venus's surface pressure was 75-100 atmospheres, much higher than Venera 4's 25 atm hull strength, and the claim was retracted.
Realizing the ships would be crushed before reaching the surface, the Soviets launched Venera 5 and Venera 6 as atmospheric probes. Designed to jettison nearly half their payload prior to entering the planet's atmosphere, these craft recorded 53 and 51 minutes of data, respectively, while slowly descending by parachute before their batteries failed. This represented a significant scientific accomplishment.
It had become increasingly clear that Venus was unlikely to have liquid bodies of water but designs for the Soviet Venera probes still considered the possibility of a water landing as late as 1964. The Venera 7 probe, launched in August 1970, was the first one designed to survive Venus's surface conditions and to make a soft landing. Massively overbuilt to ensure survival, it had few experiments on board, and scientific output from the mission was further limited due to an internal switchboard failure that stuck in the "transmit temperature" position. Still, the control scientists succeeded in extrapolating the pressure (90 atm) from the temperature data with 465 °C (869 °F), which resulted from the first direct surface measurements.
The Doppler measurements of the Venera 4 to 7 probes were the first evidence of the existence of zonal winds with high speeds of up to 100 metres per second (220 mph) in the Venusian atmosphere (super rotation). Along with the pressure and temperature data acquired Venera 7 also measured atmospheric composition. Venera 7's parachute failed shortly before landing very close to the surface.
It impacted at 17 metres per second (56 ft/s) and toppled over, but survived.
This caused antenna misalignment making the radio signal very weak, but it was detected (with temperature telemetry) for 23 more minutes before its batteries expired. Thus, it became, on 15 December 1970, the first human-made probe to transmit data from the surface of Venus. Venera 8, launched in 1972, was equipped with an extended set of scientific instruments for studying the surface (gamma-spectrometer etc.).
The cruise bus of Venera 7 and 8 was similar to that of earlier ones, with the design ascending to the Zond 3 mission. The lander transmitted data during the descent and landed in sunlight. It measured the light level but had no camera. It transmitted data for almost an hour. Following the failed Kosmos 482, the 1975 Venera 9 and 10 probes and 1978 Venera 11 and 12 probes were of a different design.
They weighed approximately five tons and were launched by the powerful Proton booster. They included a transfer and relay bus that had engines to brake into Venus orbit and to serve as receiver and relay for the entry probe's transmissions. The entry probe was attached to the top of the bus in a spherical heat shield.

The probes were optimized for surface operations with an unusual design that included a spherical compartment to protect the electronics from atmospheric pressure and heat for as long as possible. Beneath this was a shock-absorbing "crush ring" for landing. Above the pressure sphere was a cylindrical antenna structure and a wide, dish-shaped structure that resembled an antenna but was actually an aerobrake.
They were designed to operate on the surface for a minimum of 30 minutes.
Instruments varied on different missions, but included cameras and atmospheric and soil analysis equipment. All four landers had problems with some or all of their camera lens caps not releasing. The Venera 9 lander operated for at least 53 minutes and took pictures with one of two cameras; the other lens cap did not release. These were the first pictures ever taken on the surface of another planet.
As with some of the Soviet Union's other planetary probes, the later versions were launched in pairs, with a second vehicle launched soon after the first. Venera 1 and Venera 2 were intended to fly past Venus without entering orbit. Venera 1 was launched on 12 February 1961. Telemetry on the probe failed seven days after launch. It is believed to have passed within 100,000 km (62,000 mi) of Venus and remains in heliocentric orbit.

Venera 9 (Russian: ??????-9, lit. 'Venus-9'), manufacturer's designation: 4V-1 No. 660, was a Soviet uncrewed space mission to Venus. It consisted of an orbiter and a lander. It was launched on June 8, 1975, at 02:38:00 UTC and had a mass of 4,936 kilograms (10,882 lb). The orbiter was the first spacecraft to orbit Venus, while the lander was the first to return images from the surface of another planet.
The orbiter entered Venus orbit on October 20, 1975.
A bell-shaped unit holding propulsion systems was attached to the bottom of the cylinder, and mounted on top was a 2.4-metre (7.9 ft) sphere which held the lander.
It carried instruments to take chemical and isotopic measurements, monitor the spectrum of scattered sunlight, and record electric discharges during its descent phase through the Venusian atmosphere. The spacecraft utilized a camera system, an x-ray fluorescence spectrometer, a screw drill and surface sampler, a dynamic penetrometer, and a seismometer to conduct investigations on the surface.
A parachute deployed once the lander entered the Venusian atmosphere. The parachute detached at about 50 kilometres (31 mi) above the surface; simple air braking was used for the final descent.

The Venera 13 lander was equipped with cameras to photograph the surface and spring-loaded arms to measure the compressibility of regolith. The quartz camera windows were covered by lens caps that popped off after descent. However, only one of these worked for Venera 13. The terrain at the touchdown point was bedrock outcrops breaking through dark, fine-grained regolith. After landing, the lander began to take a panoramic photograph while a mechanical drill deployed to collect a sample of the surface material.
The samples were deposited in a hermetically sealed chamber, maintained at 30 °C (86 °F) and a pressure of about 0.05 atmosphere (5 kPa). Sample composition was determined by the X-ray fluorescence spectrometer to be in the class of weakly-differentiated melanocratic alkaline gabbroids.
Although the lander was designed to function for about 32 minutes, it continued to operate for at least 127 minutes in an environment where the temperature was 457 °C (855 °F) and the pressure was 9.0 MPa (89 standard atmospheres).
The Venera lander appears in the book Contact (Sagan, 1985).
On Ksanfomality's Venus Life Result
Venera 2 (Russian: ??????-2 meaning Venus 2), also known as 3MV-4 No.4 was a Soviet spacecraft intended to explore Venus. A 3MV-4 spacecraft launched as part of the Venera programme, it failed to return data after flying past Venus.
An imaging system (cameras)
Solar and cosmic x-ray detectors
Cosmic-ray gas-discharge counters
Cosmic radio emission receivers
The spacecraft made its closest approach to Venus at 02:52 UTC on 27 February 1966, at a distance of

That streak of early failures is easy to misread from the modern side of the table.
The 3MV probes were pushing every technology of their era at once: launch vehicles barely adequate to the mass, deep-space navigation with no ground testing possible, computers that were barely more than sequencers, and thermal systems designed by analogy rather than experience. Every failed telemetry link and every dead battery produced a specific engineering response, and the program's internal culture treated each flight as a systems test as much as an attempt at science. Within five years of

these losses the same industrial base was delivering spacecraft that could survive the full descent to the surface, and within fifteen it was returning panoramas. The lesson the early Venera flights taught the rest of the field is that planetary programs are marathons run in laps, and the first laps are usually ugly. What distinguishes a program that eventually lands is not a clean record but the willingness to launch the correction. 23,810 km (14,790 mi). During the flyby, all of Venera 2's instruments were activated, requiring that radio contact with the spacecraft be suspended.